Faculty of Computer Science and Mathematics
Universiti Malaysia Terengganu

LAB REPORT

CSE3023 Web-Based Application Development

Name	Wai Pei Yuen
Matric No.	S74991
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Lab Report	2
Demonstrator	Ts. Dr. Wan Nural Jawahir Hj Wan Yussof

Task 1:

Reflection:

1. What have you learned from this exercise?

I understand the flow: user submits form → LognServlet processes → forwards to AccountServlet → display shared data.

2. What are the common methods used in Java Servlet?

doPost(HttpServletRequest request, HttpServletResponse response)

-handles post requests from the login form.

RequestDispatcher.forward.forward(request, response)

- forward request to another servlet.

Exercise:

Source Code:

add_product.html

```
<!DOCTYPE html>
<!--
Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
Click nbfs://nbhost/SystemFileSystem/Templates/JSP Servlet/Html.html to edit this template
-->
<html>
    <head>
        <title>Add Product</title>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
    </head>
    <body>
        <h2>Product Data</h2>
        <form action="InsertProductServlet" method="POST">
            <label for="name">Name:</label>
            <input type="text" id="name" name="name" required><br><br>

            <label for="category">Category:</label>
            <input type="text" id="category" name="category" required><br><br>

            <label for="price">Price:</label>
            <input type="number" id="price" step="0.01" name="price" required><br><br>

            <label for="quantity">Quantity:</label>
            <input type="number" id="quantity" name="quantity" required><br><br>

            <input type="submit" value="Add Product">
        </form>
    </body>
</html>
```

Product.java

```java
//User JavaBean Clas
package com.lab.model;

public class Product {
    private int id;
    private String name;
    private String category;
    private double price;
    private int quantity;

    // Constructor Kosong (Wajib untuk JavaBean)
        public Product() {
    }

    // Constructor dengan ID (Untuk Update & Delete)
    public Product(int id, String name, String category, double price, int quantity) {
        this.id = id;
        this.name = name;
        this.category = category;
        this.price = price;
        this.quantity = quantity;
    }

    // Constructor tanpa ID (Untuk Insert)
    public Product(String name, String category, double price, int quantity) {
        this.name = name;
        this.category = category;
        this.price = price;
        this.quantity = quantity;
    }

    // Getter dan Setter
    public int getId() {
        return id;
    }

    public void setId(int id) {
        this.id = id;
    }
```

```java
    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }

    public String getCategory() {
        return category;
    }

    public void setCategory(String category) {
        this.category = category;
    }

    public double getPrice() {
        return price;
    }

    public void setPrice(double price) {
        this.price = price;
    }

    public int getQuantity() {
        return quantity;
    }

    public void setQuantity(int quantity) {
        this.quantity = quantity;
    }
}
```

ProductDAO.java

```java
//class userDAO
package com.lab.dao;

import com.lab.model.Product;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.util.ArrayList;
import java.util.List;

public class ProductDAO {

    private String jdbcURL = "jdbc:mysql://localhost:3307/CSE3023";
    private String jdbcUsername = "root";
    private String jdbcPassword = "";

    // Method untuk mendapatkan sambungan Database
    protected Connection getConnection() {
        Connection connection = null;
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            connection = DriverManager.getConnection(jdbcURL,
                    jdbcUsername, jdbcPassword);
        } catch (SQLException | ClassNotFoundException e) {
            e.printStackTrace();
        }
        return connection;
    }

    // INSERT
    public void insertProduct(Product product) {
        String sql = "INSERT INTO products (name, category, price, quantity) VALUES( ?,  ?,  ?, ?)";

        try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(sql)) {
            pstmt.setString(1, product.getName());
            pstmt.setString(2, product.getCategory());
            pstmt.setDouble(3, product.getPrice());
            pstmt.setInt(4, product.getQuantity());
```

```java
    try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(sql)) {
        pstmt.setString(1, product.getName());
        pstmt.setString(2, product.getCategory());
        pstmt.setDouble(3, product.getPrice());
        pstmt.setInt(4, product.getQuantity());

        pstmt.executeUpdate();
    } catch (SQLException e) {
        e.printStackTrace();
    }
}

// SELECT ALL
public List<Product> selectAllProducts() {
    List<Product> list = new ArrayList<>();
    String sql = "SELECT * FROM products";

    try (Connection conn = getConnection();
         PreparedStatement ps = conn.prepareStatement(sql);
         ResultSet rs = ps.executeQuery()) {

        while (rs.next()) {
            list.add(new Product(
                    rs.getInt("id"),
                    rs.getString("name"),
                    rs.getString("category"),
                    rs.getDouble("price"),
                    rs.getInt("quantity")
            ));
        }

    } catch (Exception e) {
        e.printStackTrace();
    }
    return list;
}
```

```java
// SELECT BY ID
public Product selectProduct(int id) {
    Product p = null;
    String sql = "SELECT * FROM products WHERE id=?";

    try (Connection conn = getConnection();
         PreparedStatement ps = conn.prepareStatement(sql)) {

        ps.setInt(1, id);
        ResultSet rs = ps.executeQuery();

        if (rs.next()) {
            p = new Product(
                    id,
                    rs.getString("name"),
                    rs.getString("category"),
                    rs.getDouble("price"),
                    rs.getInt("quantity")
            );
        }

    } catch (Exception e) {
        e.printStackTrace();
    }
    return p;
}


// UPDATE
public boolean updateProduct(Product p) {
    boolean rowUpdated = false;
    String sql = "UPDATE products SET name=?, category=?, price=?, quantity=? WHERE id=?";

    try (Connection conn = getConnection();
         PreparedStatement ps = conn.prepareStatement(sql)) {

        ps.setString(1, p.getName());
        ps.setString(2, p.getCategory());
        ps.setDouble(3, p.getPrice());
        ps.setInt(4, p.getQuantity());
        ps.setInt(5, p.getId());
```

```java
    // UPDATE
    public boolean updateProduct(Product p) {
        boolean rowUpdated = false;
        String sql = "UPDATE products SET name=?, category=?, price=?, quantity=? WHERE id=?";

        try (Connection conn = getConnection();
             PreparedStatement ps = conn.prepareStatement(sql)) {

            ps.setString(1, p.getName());
            ps.setString(2, p.getCategory());
            ps.setDouble(3, p.getPrice());
            ps.setInt(4, p.getQuantity());
            ps.setInt(5, p.getId());

            rowUpdated = ps.executeUpdate() > 0;

        } catch (Exception e) {
            e.printStackTrace();
        }
        return rowUpdated;
    }

    // DELETE
    public boolean deleteProduct(int id) {
        boolean rowDeleted = false;
        String sql = "DELETE FROM products WHERE id=?";

        try (Connection conn = getConnection();
             PreparedStatement ps = conn.prepareStatement(sql)) {

            ps.setInt(1, id);
            rowDeleted = ps.executeUpdate() > 0;

        } catch (Exception e) {
            e.printStackTrace();
        }
        return rowDeleted;
    }
}
```

InsertProductServlet.java

```java
package com.lab.controller;

import com.lab.dao.ProductDAO;
import com.lab.model.Product;
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;


/**
 *
 * @author wpy92
 */
public class InsertProductServlet extends HttpServlet {

    /**
     * Processes requests for both HTTP <code>GET</code> and <code>POST</code>
     * methods.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */

    private ProductDAO dao;

    @Override
    public void init() {
        dao = new ProductDAO();
    }

    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        try (PrintWriter out = response.getWriter()) {
            /* TODO output your page here. You may use following sample code. */
            out.println("<!DOCTYPE html>");
            out.println("<html>");
```

```java
    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        try (PrintWriter out = response.getWriter()) {
            /* TODO output your page here. You may use following sample code. */
            out.println("<!DOCTYPE html>");
            out.println("<html>");
            out.println("<head>");
            out.println("<title>Servlet InsertServlet</title>");
            out.println("</head>");
            out.println("<body>");
            out.println("<h1>Servlet InsertServlet at " + request.getContextPath() + "</h1>");
            out.println("</body>");
            out.println("</html>");
        }
    }

    // <editor-fold defaultstate="collapsed" desc="HttpServlet methods. Click on the + sign on the left to edit the code.">
    /**
     * Handles the HTTP <code>GET</code> method.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
```

```java
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }

    /**
     * Handles the HTTP <code>POST</code> method.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        String name = request.getParameter("name");
        String category = request.getParameter("category");
        double price = Double.parseDouble(request.getParameter("price"));
        int quantity = Integer.parseInt(request.getParameter("quantity"));

        Product newProduct = new Product(name, category, price, quantity);
        dao.insertProduct(newProduct); // Panggil DAO, bukan JDBC
        response.sendRedirect("ViewProductServlet");

    }

    /**
     * Returns a short description of the servlet.
     *
     * @return a String containing servlet description
     */
    @Override
    public String getServletInfo() {
        return "Short description";
    }// </editor-fold>

}
```

ViewProductServlet.java

```java
package com.lab.controller;

import com.lab.dao.ProductDAO;
import com.lab.model.Product;
import java.io.IOException;
import java.io.PrintWriter;
import java.util.List;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

public class ViewProductServlet extends HttpServlet {

    private ProductDAO dao;

    @Override
    public void init() {
        dao = new ProductDAO();
    }

@Override
protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

    response.setContentType("text/html");
    PrintWriter out = response.getWriter();

    List<Product> list = dao.selectAllProducts();

    out.println("<h2>Product List</h2>");
    out.println("<table border='1'>");
    out.println("<tr><th>ID</th><th>Name</th><th>Category</th><th>Price</th><th>Qty</th><th>Action</th></tr>");

    for (Product p : list) {
        out.println("<tr>");
        out.println("<td>" + p.getId() + "</td>");
        out.println("<td>" + p.getName() + "</td>");
        out.println("<td>" + p.getCategory() + "</td>");
        out.println("<td>" + p.getPrice() + "</td>");
        out.println("<td>" + p.getQuantity() + "</td>");
        out.println("<td>");
        out.println("<a href='UpdateProductServlet?id=" + p.getId() + "'>Edit</a> | ");
        out.println("<a href='DeleteProductServlet?id=" + p.getId() + "'>Delete</a>");
        out.println("</td>");
        out.println("</tr>");
    }

    out.println("</table>");
    out.println("<br><a href='add_product.html'>Add Product</a>");
}

    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        doGet(request, response); // Redirect POST to GET
    }

    @Override
    public String getServletInfo() {
        return "ViewServlet displays all users";
    }
}
```

DeleteProductServlet.java

```java
package com.lab.controller;

/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/JSP Servlet/Servlet.java to edit this template
 */
import com.lab.dao.ProductDAO;
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

/**
 *
 * @author wpy92
 */
public class DeleteProductServlet extends HttpServlet {

    /**
     * Processes requests for both HTTP <code>GET</code> and <code>POST</code>
     * methods.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    private ProductDAO dao;

    @Override
    public void init() {
        dao = new ProductDAO();
    }

protected void processRequest(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
    response.setContentType("text/html;charset=UTF-8");
    try (PrintWriter out = response.getWriter()) {
        /* TODO output your page here. You may use following sample code. */
        out.println("<!DOCTYPE html>");
        out.println("<html>");
        out.println("<head>");
        out.println("<title>Servlet DeleteServlet</title>");
        out.println("</head>");
        out.println("<body>");
        out.println("<h1>Servlet DeleteServlet at " + request.getContextPath() + "</h1>");
        out.println("</body>");
        out.println("</html>");
    }
}

// <editor-fold defaultstate="collapsed" desc="HttpServlet methods. Click on the + sign on the left to edit the code.">
/**
 * Handles the HTTP <code>GET</code> method.
 *
 * @param request servlet request
 * @param response servlet response
 * @throws ServletException if a servlet-specific error occurs
 * @throws IOException if an I/O error occurs
 */
```

```java
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response) throws ServletException, IOException {

        int id = Integer.parseInt(request.getParameter("id"));
        dao.deleteProduct(id); // Panggil DAO

        response.sendRedirect("ViewProductServlet");
    }

    /**
     * Handles the HTTP <code>POST</code> method.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }

    /**
     * Returns a short description of the servlet.
     *
     * @return a String containing servlet description
     */
    @Override
    public String getServletInfo() {
        return "Short description";
    }// </editor-fold>

}
```

UpdateProductServlet.java

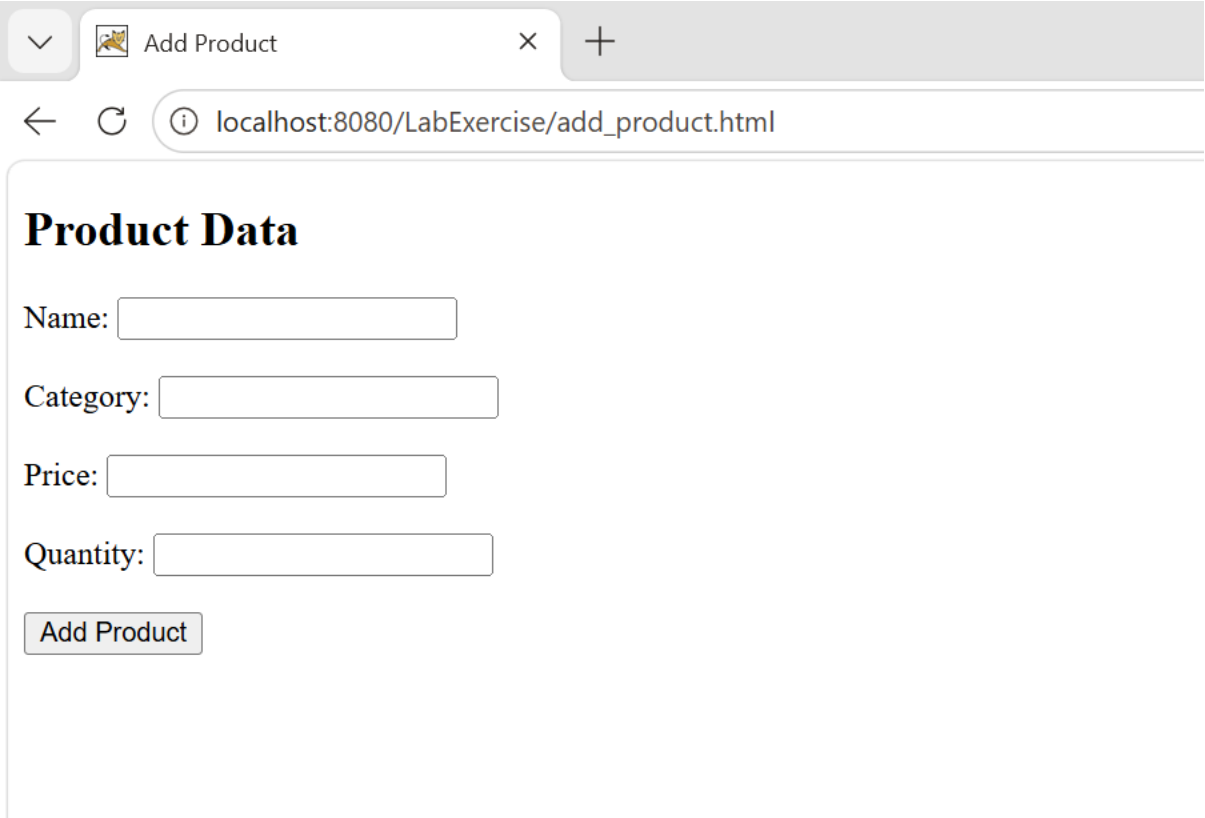
```java
package com.lab.controller;

/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/JSP Servlet/Servlet.java to edit this template
 */
import com.lab.dao.ProductDAO;
import com.lab.model.Product;
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;


/**
 *
 * @author wpy92
 */
public class UpdateProductServlet extends HttpServlet {

    /**
     * Processes requests for both HTTP <code>GET</code> and <code>POST</code>
     * methods.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    private ProductDAO dao;

    @Override
    public void init() {
        dao = new ProductDAO();
    }
```

```java
    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        try (PrintWriter out = response.getWriter()) {
            /* TODO output your page here. You may use following sample code. */
            out.println("<!DOCTYPE html>");
            out.println("<html>");
            out.println("<head>");
            out.println("<title>Servlet UpdateServlet</title>");
            out.println("</head>");
            out.println("<body>");
            out.println("<h1>Servlet UpdateServlet at " + request.getContextPath() + "</h1>");
            out.println("</body>");
            out.println("</html>");
        }
    }

    // <editor-fold defaultstate="collapsed" desc="HttpServlet methods. Click on the + sign on the left to edit the code.">
    /**
     * Handles the HTTP <code>GET</code> method.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
```

```java
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        int id = Integer.parseInt(request.getParameter("id"));
        Product product = dao.selectProduct(id); // Call DAO

        if (product != null) {
            out.println("<h2>Update Product</h2>");
            out.println("<form action='UpdateProductServlet' method='POST'>");

            out.println("<input type='hidden' name='id' value='" + product.getId() + "'>");

            out.println("Name: <input type='text' name='name' value='" + product.getName() + "' required><br><br>");
            out.println("Category: <input type='text' name='category' value='" + product.getCategory() + "' required><br><br>");
            out.println("Price: <input type='number' step='0.01' name='price' value='" + product.getPrice() + "' required><br><br>");
            out.println("Quantity: <input type='number' name='quantity' value='" + product.getQuantity() + "' required><br><br>");

            out.println("<input type='submit' value='Update Product'>");
            out.println("</form>");
        } else {
            out.println("<h3>Product not found!</h3>");
        }
    }

    /**
     * Handles the HTTP <code>POST</code> method.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
```

```java
    // Simpan data yang telah dikemaskini
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        int id = Integer.parseInt(request.getParameter("id"));
        String name = request.getParameter("name");
        String category = request.getParameter("category");
        double price = Double.parseDouble(request.getParameter("price"));
        int quantity = Integer.parseInt(request.getParameter("quantity"));

        Product product = new Product(id, name, category, price, quantity);

        dao.updateProduct(product); // Call DAO (NO JDBC here)

        response.sendRedirect("ViewProductServlet");
    }

    /**
     * Returns a short description of the servlet.
     *
     * @return a String containing servlet description
     */
    @Override
    public String getServletInfo() {
        return "Short description";
    }// </editor-fold>

}
```

Input:


Add Product
localhost:8080/LabExercise/add_product.html
Product Data
Name:
Category:
Price:
Quantity:
Add Product

Output:

Product List

ID	Name	Category	Price	Qty	Action
1	1	1	1.0	1	Edit | Delete
2	2	2	2.0	2	Edit | Delete

Add Product

Edit product:

Update Product

Name: 1

Category: 1

Price: 1.0

Quantity: 1

Update Product

Delete Product:

Product List

ID	Name	Category	Price	Qty	Action
1	1	1	1.0	1	Edit | Delete

Add Product